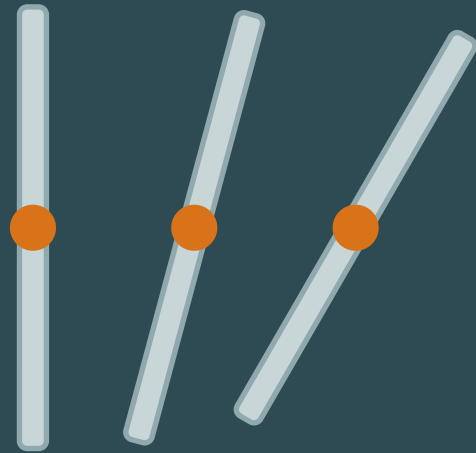
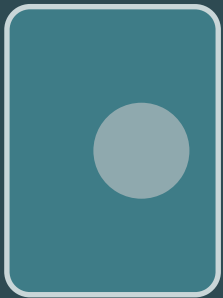
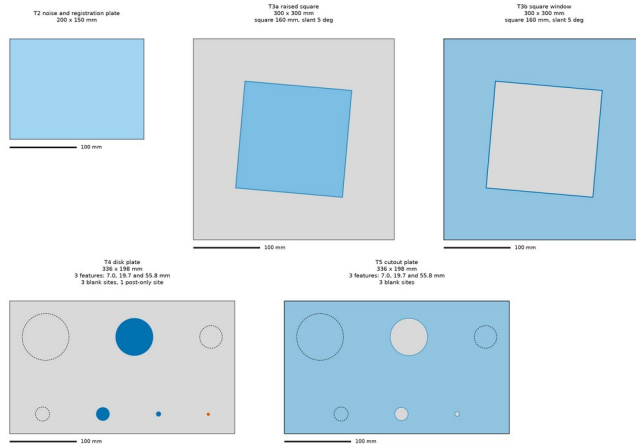


VSX3000 Performance Testing: Procurement and Build

The targets, the mounting system and the fixtures needed for the characterization captures



Five targets, one two-plane construction



Front views of T2, T3a, T3b, T4 and T5 to one scale, from the code's own target definitions at the indicative sensor geometry. Gray is the back plate, dashed circles are blank sites, the dot on T4 is the post-only site.



T2, the board

The stage-1 calibration board, 200 x 150 mm, flat to 0.05 mm, on its own board adapter: registration, noise, depth steps and the drift sentinels



T3a and T3b, the edge targets

A raised 160 mm square on hidden standoffs, and a 160 mm window in a front plate; 300 x 300 mm back plates; both edge polarities for lateral resolution



T4 and T5, the feature plates

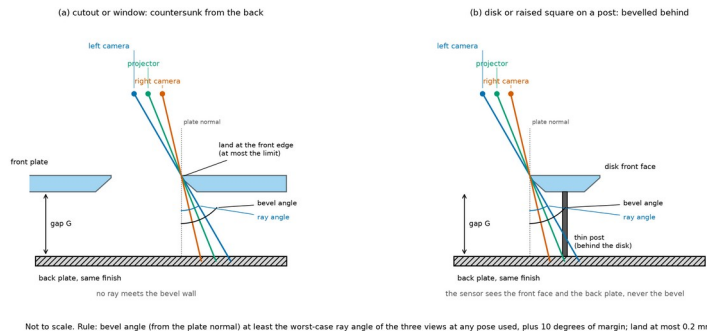
Three disks of 7.0, 19.7 and 55.8 mm on 2 mm posts, and three holes of the same sizes in a front plate; 336 x 198 mm; blank sites give the false-alarm rate



Two gaps

Standoffs set the gap G, 15 mm or 60 mm, between the faces the sensor sees, so every result is measured at two step heights

Every boundary is a knife edge beveled from the back



Cross-section of a cutout and a disk on its post, not to scale. The three viewpoints pass the knife edge in open space and never meet the beveled wall.



Bevel

45 degrees from the plate normal, cut from the back: holes and the window widen away from the sensor, disks and the square narrow away from it



Land

The flat land at the front edge is 0.2 mm or less, dimensioned 0.1 plus or minus 0.1 mm



Posts

Disks stand on 2 mm stainless posts, thinner than half the expected minimum detectable size; a post-only site shows whether the post itself is seen



One finish

Fine glass-bead blast on every face the sensor sees, all plates in one batch, so no reflectance difference biases an edge

What must be built

Item	Quantity	Drawing	From
Target adapter: flange plate with the spigot bore and the cross-pin hole	1	PT-01	new
Target spigots, one per feature target	4	PT-02	new
Standoff sets, 15 mm and 60 mm gaps	4 lengths, 32 pieces with spares	PT-03	new
T3a, raised square	1	PT-04	new
T3b, square window	1	PT-05	new
T4, disk plate	1	PT-06	new
T5, cutout plate with its edge bracket	1	PT-07	new
Noise and registration board (T2) with its adapter	1	SC1-05	stage 1
Run-out fixture	1	none needed	stage 1
Rigid sensor mount	1	existing drawing	stage 1



Section 1b of the procedure. The stage-1 items exist if the stage-1 calibration project has been run; otherwise they are built to the stage-1 drawing and specification.

What must be bought

Item	Purpose	Estimated cost (USD)	From
Ball-lock pins, 8 mm, 2 off	Hold a target's spigot in the adapter; one spare	\$40 to \$120	new
Calipers, 600 mm, or a 500 mm steel rule	Plate width and length, the in-house size check	\$40 to \$250	new
Calipers with depth rod, 150 mm	Gap G and standoff lengths	\$30 to \$150	stage 1
Torque wrench	Recorded adapter and spigot screw torques	\$80 to \$250	stage 1
Temperature loggers, 2	Sensor housing and air, one sample per minute	\$60 to \$200	new
Consumables	Thread locker, retaining compound for the disk posts, isopropyl alcohol, padded cases	\$40 to \$120	new

A phone camera is enough for the setup photos. Already in hand: the sensor on its rigid mount, the robot, the capture computer with the VSX3000 SDK and the robot interface, the analysis computer.

New spending: \$2,470 to \$6,540

Item to build	Estimated cost (USD)
Target adapter (PT-01)	\$300 to \$700
Target spigots, 4 (PT-02)	\$240 to \$600
Standoff sets (PT-03)	\$150 to \$450
T3a, raised square (PT-04)	\$350 to \$900
T3b, square window (PT-05)	\$350 to \$900
T4, disk plate (PT-06)	\$450 to \$1,200
T5, cutout plate (PT-07)	\$450 to \$1,100

\$2,470 to \$6,540

new items: \$2,290 to \$5,850 to build, \$180 to \$690 to buy

\$3,120 to \$8,510

if stage 1 has not been run: its board, adapter, run-out fixture, calipers and torque wrench added

Estimates of October 2026 from suppliers' list prices and job-shop rates of \$80 to \$150 per hour, plus or minus 50 percent; the four two-plane targets are the least certain figures.

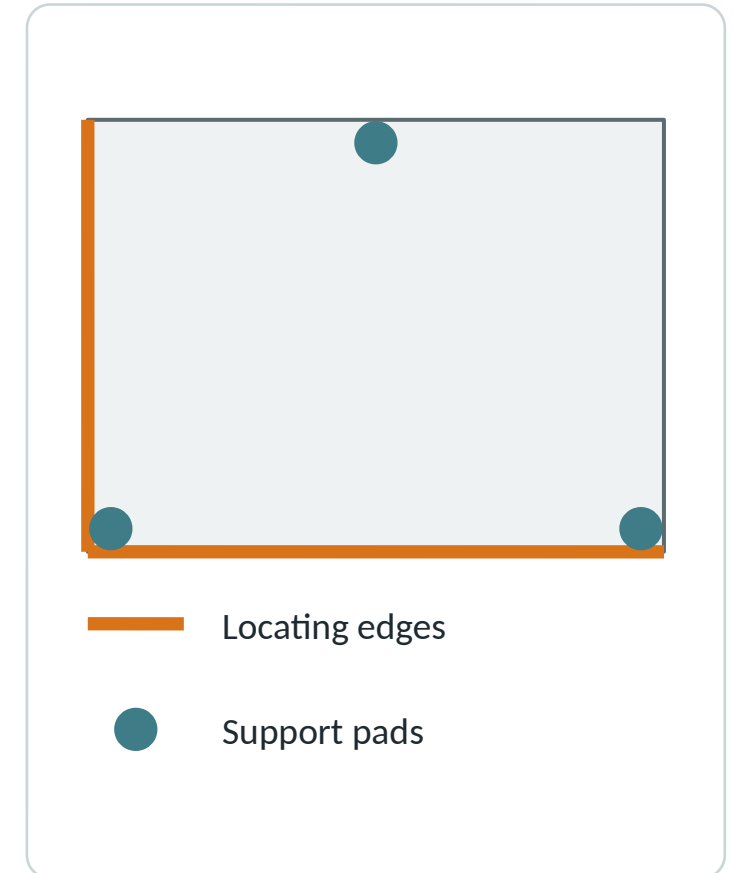
Purchase specification: the four feature targets

Requirement	T3a and T3b (edge targets)	T4 and T5 (feature plates)
Plates	300 x 300 mm; the raised square and the window 160 mm, slanted 5 degrees	336 x 198 mm; disks and holes of 7.0, 19.7 and 55.8 mm; three blank sites each, one post-only site on T4
Thickness	Front plate 6 mm, back plates 8 mm, raised square 10 mm	Front plate 6 mm, back plates 8 mm; disks 2, 3 and 4 mm
Knife edges	Beveled or countersunk from the back at 45 degrees; land 0.2 mm or less	same
Sizes known to	The fabricator's inspection report: size, land, bevel, position from the spigot datum, each with its uncertainty	same
Flatness	0.05 mm over each front face, checked in-house with the run-out fixture after mounting	same
Material and finish	Finish-ground aluminum tooling plate; fine glass-bead blast on every face the sensor sees, all plates in one batch	same; disks turned from aluminum on 2 mm stainless posts
Mounting	Spigot PT-02 on the back plate: four M5 on a 60 mm pitch circle; standoffs PT-03 for both gaps	same; T5's spigot sits on an edge bracket so the back plate can be removed

Procedure section 1d, Table 3; hand it to the fabricator with the drawings of appendix F. The four targets are one order so that every surface has the same finish.

Purchase specification: the board (from stage 1)

Requirement	Noise and registration board (T2)
Size	200 x 150 mm, at least 6 mm thick
Material	Finish-ground aluminum tooling plate (for example MIC-6) or float glass
Flatness	0.05 mm over the front face, checked in-house with the run-out fixture after mounting on the board adapter
Front face	Thin, even matte light-gray paint, or bead blast if the plate is 10 mm or thicker; the same mid-range reflectance as the feature targets
Edges	Bottom long edge and left short edge square and clean: they rest on the adapter's edge pins
Documents	Supplier's flatness report, kept with the session notes



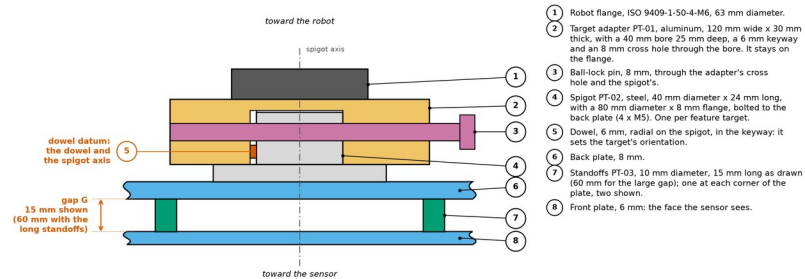
Procedure section 1d, Table 4: the stage-1 specification repeated. The board stays on its SC1-05 adapter; it never uses the spigot.

Candidate suppliers

Item	Candidate suppliers	Notes
Plate stock and machining	McMaster-Carr MIC-6 plate; any job shop with a CNC mill, a lathe and bead blasting	Finish-grind the front faces; ask for the inspection report with the quote
Posts and spigot stock	McMaster-Carr, MSC Industrial	2 mm stainless drill rod; precision ground bar for the spigots
Ball-lock pins	McMaster-Carr, Carr Lane, Jergens	8 mm, T-handle, grip length as on PT-01; about \$20 to \$60 each
Temperature loggers	Lascar EL-USB-1, Onset HOBO MX100	0.1 degree C resolution, CSV export; about \$30 to \$100 each
Measuring instruments	Mitutoyo, Starrett, iGaging; McMaster-Carr, Transcat, MSI-Viking	600 mm calipers or a 500 mm steel rule; the stage-1 calipers, dial indicator, magnetic base and torque wrench
Board	McMaster-Carr MIC-6 plus a grinding shop; Edmund Optics float glass	Stage 1; flatness report, matte paint, re-checked after mounting

From suppliers' web pages, October 2026; not an endorsement. Confirm sizes, finishes and prices before ordering.

One adapter, one spigot per target



The mounting stack in cross-section, not to scale: robot flange, adapter, spigot with its dowel in the keyway, ball-lock pin, back plate, standoffs, front plate.



Target adapter PT-01

Aluminum, 120 x 120 x 30 mm, on the ISO 9409-1-50-4-M6 flange; 40 mm H7 bore 25 mm deep with a keyway and an 8 mm cross hole



Spigot PT-02

Stainless, 40 mm h6 body, 80 mm flange bolted to the target's back plate with four M5; a radial dowel sets the orientation



Ball-lock pin

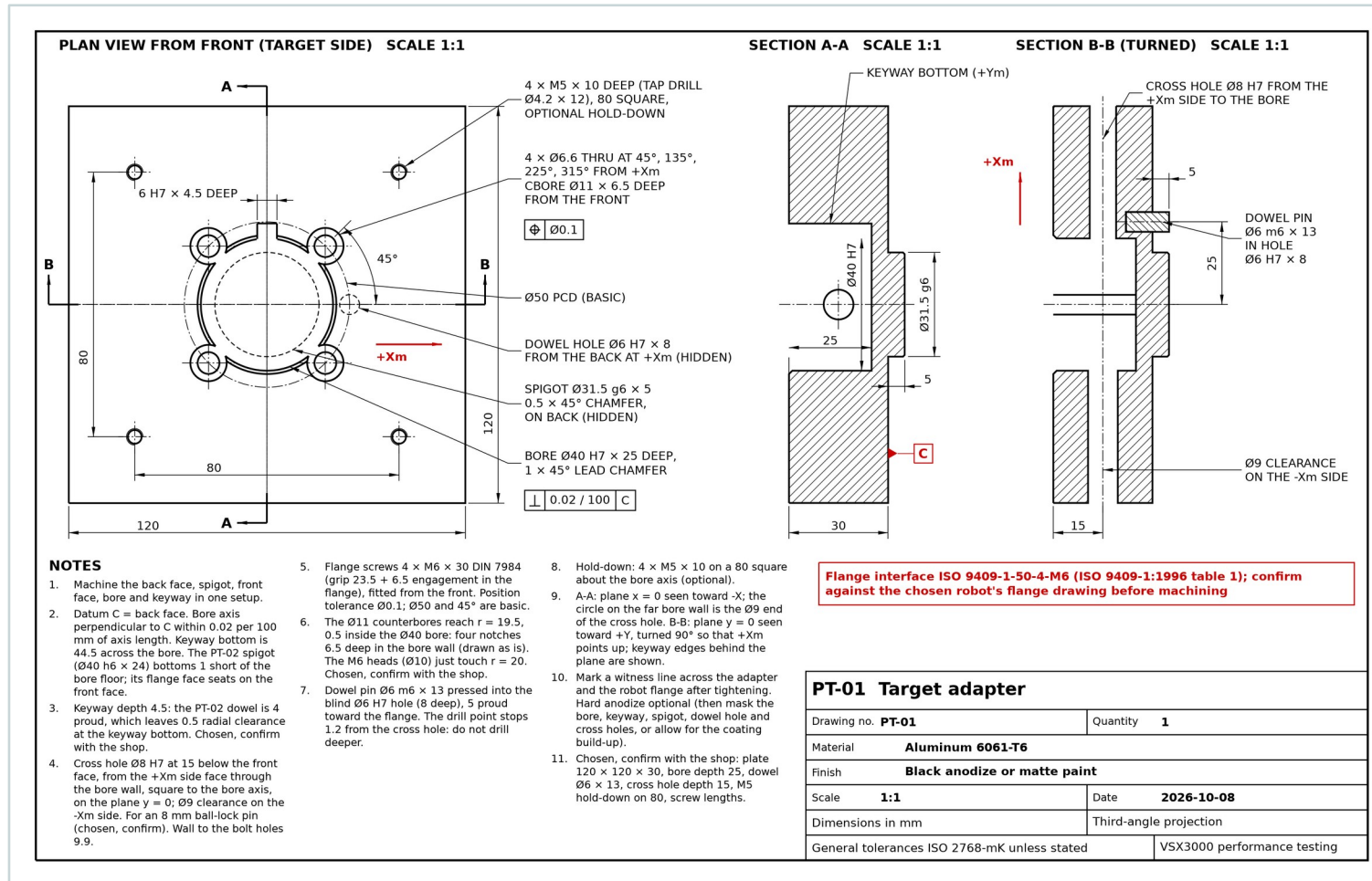
Through the adapter's cross hole and the spigot's: pull it, lift the target off, drop the next one in



The datum

The spigot axis and the dowel are the datum every as-built offset refers to; the once-per-mount check confirms each mount

Target adapter PT-01



Aluminum 6061-T6, 120 x 120 x 30 mm, machined in one setup: back face, spigot, front face, bore and keyway



ISO 9409-1-50-4-M6 side: 31.5 mm g6 spigot, four M6 x 30 on the 50 mm pitch circle, 6 mm dowel at +Xm; confirm against the robot's flange drawing



Bore 40 mm H7, 25 mm deep, with a 6 mm keyway 4.5 mm deep for the spigot's dowel

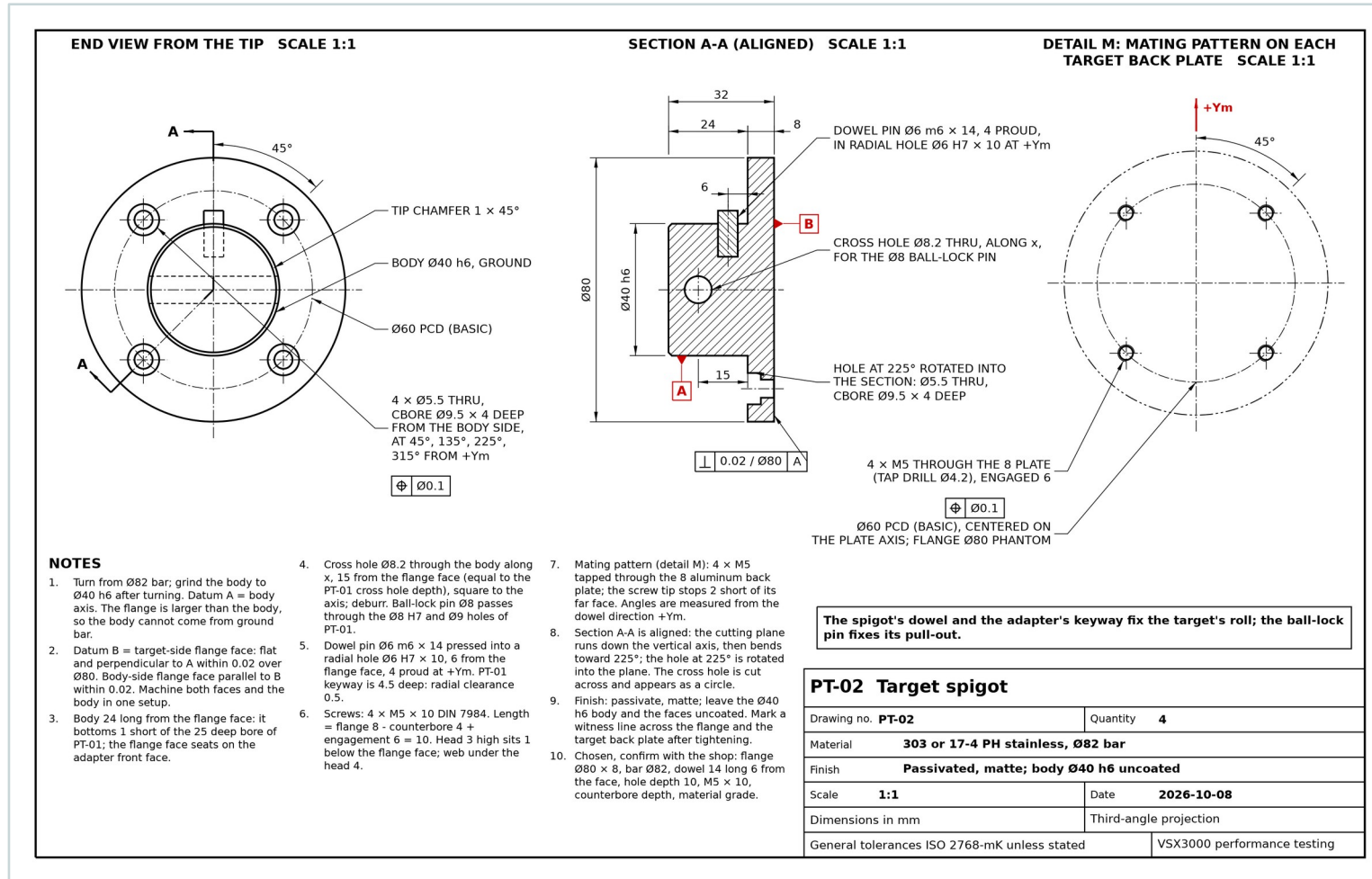


Cross hole 8 mm H7 from the +Xm side, 15 mm below the front face, for the ball-lock pin; 9 mm clearance on the far side



Bore axis square to the back face within 0.02 mm per 100 mm; hard anodize optional

Target spigot PT-02 and its mating pattern



303 or 17-4 PH stainless, turned from 82 mm bar; body 40 mm h6, ground, 24 mm long; flange 80 x 8 mm



Dowel pin 6 mm, 4 mm proud at +Ym, in a radial hole 6 mm from the flange face: it fixes the target's roll



Cross hole 8.2 mm along x, 15 mm from the flange face, for the ball-lock pin

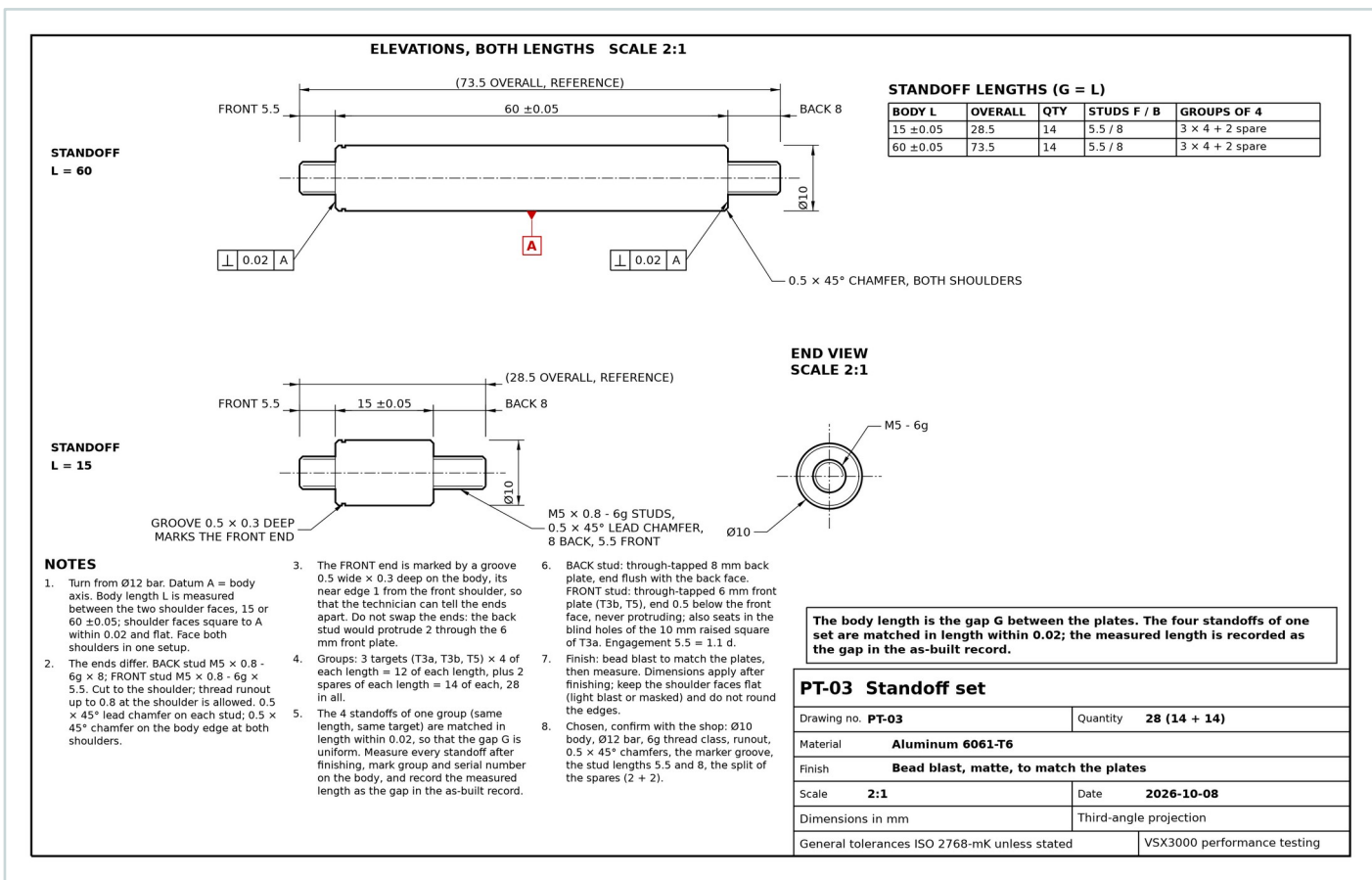


Four M5 x 10 screws on a 60 mm pitch circle at 45, 135, 225 and 315 degrees from the dowel, counterbored from the body side



Every target's back plate carries the same pattern: four M5 tapped through the 8 mm plate, centered on the plate axis

Standoff set PT-03: the body length is the gap



Aluminum, 10 mm body; the body length is G minus the front part's thickness, so the step the sensor sees is G: 9 and 54 mm for the 6 mm front plates, 5 and 50 mm for the 10 mm raised square, plus or minus 0.05 mm



Asymmetric studs: M5 x 8 at the back end into the 8 mm back plate, M5 x 5.5 at the front end so it sits 0.5 mm below a 6 mm front face; a groove marks the front end



Four standoffs of one group are matched in length within 0.02 mm; the measured length is recorded as the gap in the as-built record

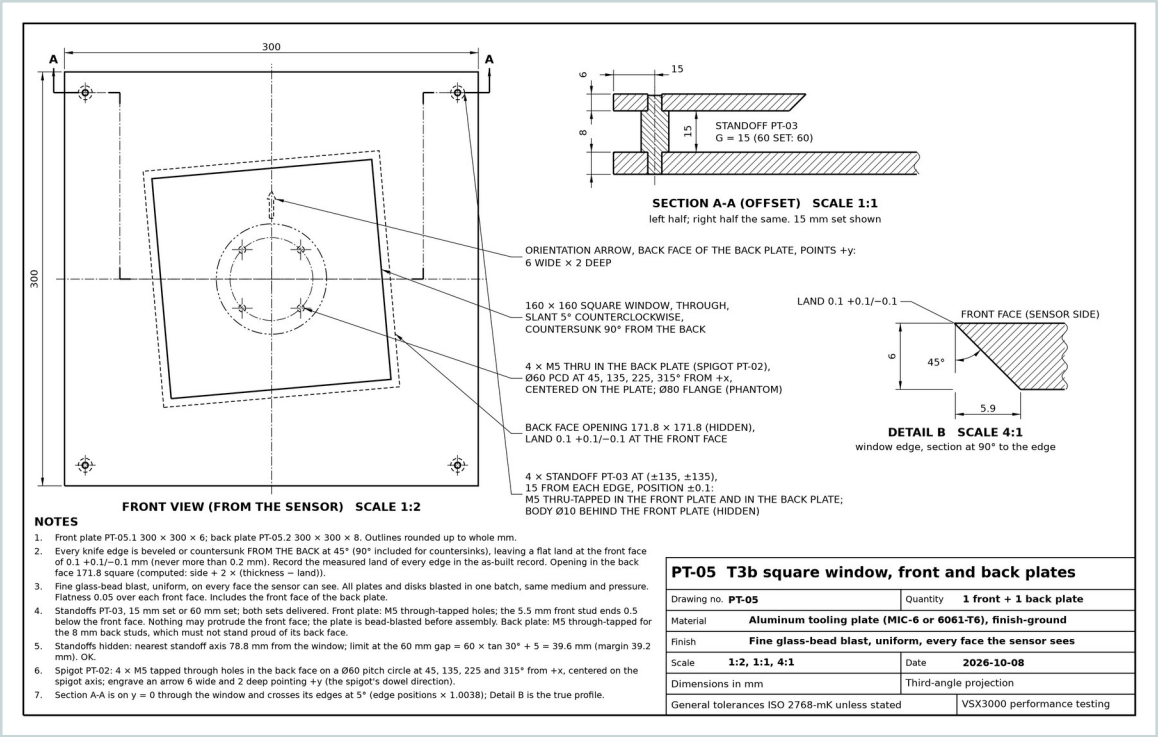
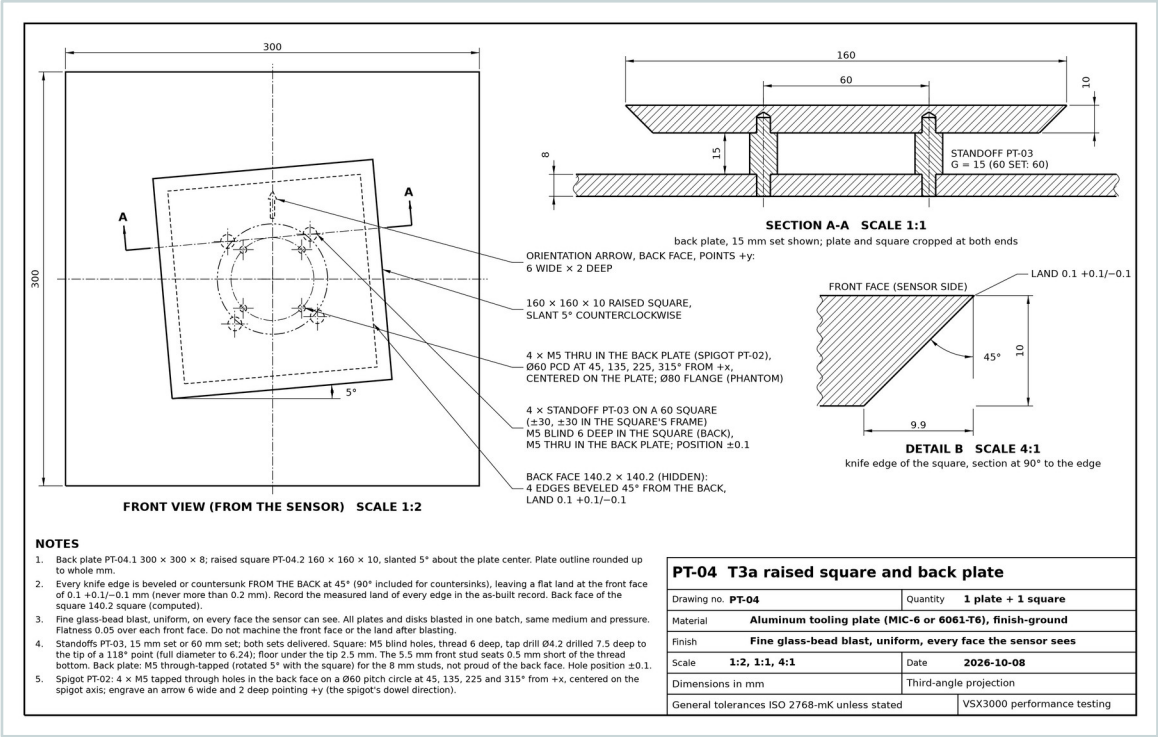


Three targets use them: hidden behind the raised square of T3a, at the corners of T3b and T5; 32 pieces with spares



Bead blast to match the plates, then measure

Edge targets PT-04 and PT-05



Left: T3a, the raised 160 mm square on four hidden standoffs above a 300 mm back plate. Right: T3b, the 160 mm window countersunk from the back of a 300 mm front plate, standoffs at the corners.

FRONT VIEW (FROM THE SENSOR) SCALE 1:2

SITES (target frame, origin at plate center); POSITION TOL ±0.1

Site	Kind	x	y	Front Ø	Feature in plate
disk_00	disk	+54.19	+69.75	6.98 ±0.02	Ø2 H7 x 6 blind
disk_01	disk	-28.92	+69.75	19.73 ±0.02	Ø2 H7 x 6 blind
disk_02	disk	+18.04	-45.34	55.80 ±0.02	Ø2 H7 x 6 blind
post_00	post	+128.44	+69.75	2 (post)	Ø2 H7 x 6 blind
blank_00	blank	-119.00	+69.75	20.93 (patch)	none: plain plate
blank_01	blank	+132.53	-45.34	33.68 (patch)	none: plain plate
blank_02	blank	-114.49	-45.34	69.75 (patch)	none: plain plate

4 × Ø2 H7 REAMED, 6 DEEP, BLIND, FROM THE FRONT FACE (3 DISK SITES + 1 POST-ONLY SITE); POSITION ±0.1
ORIENTATION ARROW, BACK FACE, POINTS +y;
6 WIDE × 2 DEEP

4 × M5 THRU IN THE BACK PLATE (SPIGOT PT-02), Ø60 PCD AT 45, 135, 225, 315° FROM +x; Ø80 FLANGE (PHANTOM)

BLANK SITE (PHANTOM CIRCLE): NO FEATURE, KEEP FREE OF MARKS (3 PLACES)

NOTES

- Back plate 336 × 198 × 8 (outline rounded up to whole mm from 335.94 × 197.63). Site positions and diameters are from the target definition (table); position ±0.1.
- Post holes Ø2 H7 reamed, 6 deep, blind, from the front face, square to it. Posts (sheet 2) are a slip fit, secured with low-strength removable retaining compound; swap the posts to change the gap. Thin circles are the disks (sheet 2); phantom circles are blank sites: no feature, keep free of marks.
- Fine glass-bead blast, uniform, on every face the sensor can see. All plates and disks blasted in one batch, same medium and pressure. Flatness 0.05 over each front face. Blast the plate face with the disks and posts in the same batch.
- Spigot PT-02: 4 × M5 tapped through holes in the back face on a Ø60 pitch circle at 45, 135, 225 and 315° from +x, centered on the spigot axis; engrave an arrow 6 wide and 2 deep pointing +y (the spigot's dowel direction).
- Gap G (15 or 60 mm) is measured from the plate's front face to the back face of the disk. The plate carries no standoffs.

SECTION A-A SCALE 1:1
through disk_02 (largest) with its post, 15 mm set

PT-06 sheet 1/2 T4 disk plate: back plate	
Drawing no: PT-06 sheet 1/2	Quantity: 1
Material: Aluminum tooling plate (MC-6 or 6061-T6), finish-ground	
Finish: Fine glass-bead blast, uniform, every face the sensor sees	
Scale: 1:2, 1:1	Date: 2026-10-08
Dimensions in mm	Third-angle projection
General tolerances ISO 2768-mK unless stated	
VSX3000 performance testing	

FRONT VIEW SCALE 1:2

SECTION A-A (OFFSET) SCALE 1:1
15 mm set; cropped

DETAIL B SCALE 4:1
cutout edge, 90° to the edge

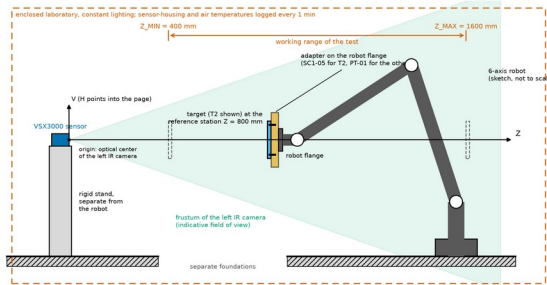
SITES (target frame); POSITION ±0.1					
Site	Kind	x	y	Front Ø ±0.02	Back Ø (CSK)
cutout_00	cutout	+90.08	+69.75	6.98	18.78
cutout_01	cutout	+6.98	+69.75	19.73	31.53
cutout_02	cutout	+18.04	-45.34	55.80	67.60
blank_00	blank	-83.10	+69.75	20.93 (patch)	none
blank_01	blank	+132.53	-45.34	33.68 (patch)	none
blank_02	blank	-114.49	-45.34	69.75 (patch)	none

NOTES

- PT-07.1 front plate 336 × 198 × 6; PT-07.2 back plate 336 × 198 × 6, plain, REMOVABLE (open-background captures), no spigot; PT-07.3 edge bracket. Outlines rounded up to whole mm; sites from the target definition (Table), position ±0.1.
- Knife edges: countersunk FROM THE BACK at 45° (90° included), land 0.1 +0.1/-0.1 mm (max 0.2); record the measured land. Blank sites: no feature, keep free of marks.
- Fine glass-bead blast, uniform, every face the sensor sees (also the front face of the bracket extension); all plates and disks blasted in one batch, same medium and pressure. Flatness 0.05 over each front face.
- Standoffs PF-03, 15 mm or 60 mm set, both delivered; four, at x = 156 and -120, y = ±87 (12 from the right and top/bottom edges). Front plate: M5 through-tapped; the 5.5 mm front stud ends 0.5 below the front face; nothing may protrude the front face; the plate is bead-blasted before assembly. Back plate: M5 through-tapped, 8 mm studs not proud of the back face.
- The bracket (8 mm) lies inside the 15 mm gap against the back of the front plate, outer face flush with the left edge; nothing sits behind the cutouts. The robot holds TS by the spigot on the extension. Screws M5 × 8 DIN 7984 into M5 through-tapped holes: engagement 4.5, tip 1.5 mm below the front face.
- Spigot PT-02 on the extension: 4 × M5 through-tapped, Ø60 PCD at 45, 135, 225, 315° from +x, centered at x = -218; arrow 6 × 2 deep, pointing +y, on the bar's back face at x = -148 (no room on the extension).

HIDDEN-STANDOFF CHECK: nearest standoff axis to a cutout edge 64.7 mm; limit at the 60 mm gap = 60 × tan 30° + 5 = 39.6 mm; margin 25.0 mm OK.

Fixtures that are bought, not drawn



The setup: the sensor on its rigid stand, the robot carrying a target on the adapter, the working range 400 to 1600 mm, the enclosed laboratory with constant lighting.



Run-out fixture

Stage 1: dial indicator, 0.01 mm graduation, on an articulating magnetic base on a steel plate bolted to the cell table; the in-house flatness check of every plate



Drift-run stand

Any rigid laboratory stand that holds the board on its adapter without creeping, for the optional 8 hour drift run with the robot idle; no drawing



Ball-lock pins

Two 8 mm quick-release pins with a T-handle, one spare, grip length as on PT-01



Temperature loggers

Two, 0.1 degree C resolution, one strapped to the sensor housing and one in the air near the target, one sample per minute, CSV export

Checks before the targets go to the robot cell

- ✓ Adapter PT-01 made to drawing; its flange interface confirmed against the chosen robot's flange drawing
- ✓ Each spigot drops into the adapter, its dowel in the keyway, and the ball-lock pin passes; no play
- ✓ Fabricator's inspection report received for every feature: size, land, bevel and position from the spigot datum, with uncertainties, entered in the as-built record
- ✓ Every plate flat to 0.05 mm after mounting, checked with the run-out fixture; width and length recorded
- ✓ Standoffs measured and grouped; the measured lengths recorded as the gaps
- ✓ Every face the sensor sees matte, uniform, the same finish on all plates; posts of T4 not detected in the post check
- ✓ Each target weighed with its spigot and the adapter, for the robot's tool load data
- ✓ Screw torques recorded; witness lines marked across the adapter and the flange

